CardioTrack CT-200

Home Blood Pressure Monitor
User Manual & Technical Reference

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1. Introduction and Intended Use

The CardioTrack CT-200 is a non-invasive automated oscillometric blood pressure monitor intended for home use by adult patients under the supervision of a healthcare professional. It measures systolic and diastolic blood pressure and pulse rate using the oscillometric method.

This device is intended for use on the upper arm of adults aged 18 years and older. It is not intended to replace clinical-grade sphygmomanometers used by trained medical staff.

1.1 Indications for Use

The CT-200 is indicated for use in the following clinical contexts:

- Home blood pressure monitoring for hypertensive patients.
- Ambulatory blood pressure tracking as part of a physician-prescribed regimen.
- Wellness monitoring for individuals at risk of cardiovascular disease.

1.2 Contraindications

Do not use the CT-200 in the following situations:

- Patients with cardiac arrhythmias, as irregular pulse may affect measurement accuracy.
- Patients with arteriovenous fistula or dialysis shunts in the measurement arm.
- During infusion or IV administration in the measurement arm.
- Patients with Raynaud's disease or other severe circulation disorders.

1.3 Patient Population

This device is validated for adult patients with an upper arm circumference between 22 cm and 42 cm. A separate paediatric cuff is available but is out of scope for this manual. Do not use on neonates or infants.

> *Note: The CT-200 has NOT been validated for use during pregnancy or in patients with pre-eclampsia. Clinical judgement should be exercised.*

2. Safety Warnings and Precautions

■ WARNING: Do not use near high-frequency surgical equipment, MRI machines, or defibrillators. Electromagnetic interference may cause inaccurate readings or device malfunction.

■ WARNING: Do not attempt to disassemble, modify, or repair the device. Unauthorized modification voids the warranty and may create electrical or mechanical hazards.

2.1 Electrical Safety

The CT-200 operates on 4 × AA alkaline batteries (IEC LR6) or via the included AC adapter (100–240 V AC, 50/60 Hz). Never use both power sources simultaneously. Disconnect the AC adapter before replacing batteries.

The device complies with IEC 60601-1 requirements for electrical safety in medical equipment. The enclosure is rated IP21 (drip-proof) but is not waterproof. Do not immerse in liquids.

2.2 Cuff Pressure Safety

The device automatically deflates if cuff pressure exceeds 300 mmHg. If over-pressurisation occurs during a measurement, the device will display error code E3 and auto-deflate within 2 seconds. Continuous re-inflation attempts without a rest period of at least 2 minutes may cause discomfort or circulatory impairment.

2.3 Electromagnetic Compatibility

The CT-200 has been tested for EMC compliance in accordance with IEC 60601-1-2 (4th edition). The device is classified as Equipment Class B, suitable for use in home and hospital environments. Portable RF communications equipment (e.g., mobile phones) should be kept at least 30 cm from the device during measurements.

2.4 Disposal and Environmental Considerations

Do not dispose of this device with household waste. Dispose of batteries in accordance with local regulations. The device contains recyclable materials; contact Tri9T Medical Devices for return and recycling instructions.

3. Device Description and Components

The CT-200 system consists of the following components:

3.1 Main Unit

The main unit houses the measurement electronics, display, and control interface. Dimensions: 135 mm × 95 mm × 55 mm. Weight (without batteries): 280 g.

The front panel includes:

- 3.5-inch LCD display (128×64 pixels) with backlight.
- START/STOP button (large, tactile, blue).
- MEMORY button for reviewing stored readings.
- CLOCK SET button for date/time configuration.
- Power indicator LED (green: on; red: low battery).

Figure 3-1: Front panel layout of the CT-200 main unit. See Appendix B for detailed diagram.

3.2 Cuff and Tubing

The standard cuff (model CT200-CF-M) fits upper arms 22–42 cm in circumference. The cuff is constructed from washable nylon fabric with a latex-free internal bladder. The air tube length is 60 cm with a standard Luer-type connector.

> *Note: Replacement cuffs must be sourced from Tri9T Medical Devices or an authorised distributor. Third-party cuffs may affect measurement accuracy and void the warranty.*

3.3 Connectivity

The CT-200 includes a micro-USB port for data export to a PC via the CardioLink software (Windows 10/11, macOS 12+). Bluetooth 5.0 LE is available for wireless data transfer to the CardioTrack mobile application (iOS 14+ / Android 9+). Data is transmitted using the Bluetooth Health Device Profile (HDP).

3.4 Memory and Data Storage

The device stores up to 120 readings (60 per user slot in a 2-user configuration). Each reading includes: date, time, systolic BP, diastolic BP, pulse rate, and an irregular heartbeat (IHB) indicator. Memory is retained when batteries are replaced, provided replacement is completed within 60 seconds.

3.4.1 Data Security and Privacy

No patient-identifiable information is stored on the device. All data exported via USB or Bluetooth is in raw measurement format (not encrypted by the device). Healthcare providers should apply appropriate data-handling policies when transferring readings.

4. Installation and Setup

4.1 Battery Installation

Open the battery compartment on the bottom of the main unit by sliding the cover in the direction indicated by the arrow. Insert 4 × AA batteries observing correct polarity (marked inside the compartment). Replace the cover until it clicks into place.

■ WARNING: Use only fresh alkaline batteries. Do not mix old and new batteries or batteries of different types. Remove batteries if the device will not be used for more than 3 months.

4.2 AC Adapter Connection

Connect the AC adapter to the DC IN port on the right side of the main unit, then plug the adapter into a standard AC outlet. The power LED will illuminate green. The battery icon on the display will be replaced by a plug icon when AC power is active.

4.3 Date and Time Setup

On first use, the display will show the time setup screen. Use the CLOCK SET button to enter setup mode, then use START/STOP to cycle through digits and MEMORY to confirm each field. Fields: Year → Month → Day → Hour → Minute. Press CLOCK SET again to exit.

4.4 Cuff Fitting

Proper cuff placement is critical for accurate readings. Follow these steps:

1. Rest for 5 minutes before measuring. Avoid caffeine, smoking, or strenuous exercise for 30 minutes prior.
2. Sit in a chair with back supported. Keep feet flat on the floor.
3. Place the left arm on a flat surface at heart level. Palm facing upward.
4. Wrap the cuff around the upper arm, 2–3 cm above the elbow. The artery marker (■) should align with the brachial artery.
5. Ensure the cuff is snug but not tight — you should be able to slide two fingers under it.
6. Connect the air tube to the main unit.

Figure 4-1: Correct cuff placement on the upper left arm.

5. Operating Instructions

5.1 Taking a Measurement

Once the device is set up (see Section 4), press the START/STOP button to begin a measurement. The cuff will automatically inflate. Remain still and do not talk during inflation and measurement. After approximately 30–60 seconds, the cuff will deflate and the reading will be displayed.

The display shows: systolic (top), diastolic (bottom), and pulse rate (right). A WHO risk indicator (colour bar) is shown if the optional display mode is enabled.

5.2 Reading the Results

Blood pressure categories per WHO/ESH 2023 guidelines:

Category	Systolic (mmHg)	Diastolic (mmHg)
Optimal	< 120	< 80
Normal	120–129	80–84
High Normal	130–139	85–89
Grade 1 HT	140–159	90–99
Grade 2 HT	160–179	100–109
Grade 3 HT	≥ 180	≥ 110

5.3 Irregular Heartbeat Detection

If the device detects an irregular heartbeat pattern during measurement, the IHB symbol (♥?) will appear on the display. This is an indication only — it does not constitute a clinical diagnosis. Consult a healthcare provider if this symbol appears frequently.

5.4 Memory Recall

Press the MEMORY button to display stored readings. The display will show the reading number, date/time, and measurement values. Readings are stored in chronological order, with the most recent first. To scroll through readings, press MEMORY repeatedly. Press START/STOP to return to normal mode.

5.5 Multi-User Mode

The CT-200 supports two user profiles (User 1 and User 2). To switch users, hold the MEMORY button for 3 seconds. Each user has an independent 60-reading memory. User profiles do not store personal identifiers — they are simply two separate memory banks.

5.5.1 Switching Users During a Measurement

Do not attempt to switch user profiles during an active measurement. If the MEMORY button is held during cuff inflation or measurement, the action will be ignored and the measurement will continue unaffected.

6. Measurement Accuracy and Clinical Validation

6.1 Accuracy Specifications

The CT-200 has been validated in accordance with ISO 81060-2:2018 (Non-invasive sphygmomanometers — Clinical validation of automated measurement type). The following accuracy specifications apply:

Parameter	Specification	Standard
Blood Pressure (BP)	±3 mmHg or 2% of reading	ISO 81060-2
Pulse Rate	±5% of reading	ISO 81060-2
Cuff Pressure Range	0–300 mmHg	IEC 60601-2-30
Static Pressure Acc.	±3 mmHg	IEC 60601-2-30

6.2 Clinical Study Results

A clinical validation study was conducted at St. Michael's Cardiovascular Institute (n=85 subjects, age range 21–78, BMI 18.5–35.2 kg/m²) under the supervision of a registered cardiologist. The study protocol was reviewed and approved by the institutional ethics board.

Study outcomes (per ISO 81060-2 grading criteria):

- Mean difference (device vs. reference): −0.8 mmHg systolic, +1.1 mmHg diastolic.
- Standard deviation of differences: 3.7 mmHg systolic, 3.4 mmHg diastolic.
- Grade A classification achieved for both systolic and diastolic measurements.

6.3 Factors Affecting Accuracy

The following factors may affect measurement accuracy and should be considered when interpreting results:

- Body movement or talking during measurement.
- Incorrect cuff size for the patient's arm circumference.
- Cuff positioned more than 2–3 cm from the elbow.
- Cold ambient temperature causing vasoconstriction.
- Cardiac arrhythmias (see Section 1.2 Contraindications).

7. Maintenance and Troubleshooting

7.1 Cleaning and Disinfection

Clean the main unit with a soft, slightly damp cloth. Do not use alcohol-based solutions, solvents, or abrasive cleaners directly on the main unit casing. The cuff fabric may be hand-washed with mild soap and water; ensure it is completely dry before use. Do not machine wash the cuff.

7.2 Storage

Store the device in the original carrying case, away from direct sunlight, extreme temperatures (below 0°C or above 45°C), and high humidity (above 85% RH). Remove batteries for storage periods exceeding 3 months.

7.3 Calibration

The CT-200 requires calibration verification every 2 years or after any significant impact or drop. Calibration must be performed by a Tri9T-authorised service centre. A calibration sticker is affixed to the underside of the main unit showing the last verification date.

7.4 Troubleshooting

Common issues and solutions:

Symptom	Possible Cause	Action
Device will not turn on	Dead/missing batteries	Replace batteries or connect AC adapter
Display shows E1	Cuff not connected	Check cuff connector and re-seat
Display shows E2	Motion artifact during reading	Repeat measurement; remain still
Display shows E3	Over-pressurisation (>300 mmHg)	Rest 2 min; repeat. If persists, contact service
Display shows E4	Cuff too loose or slipped	Re-apply cuff and repeat
Readings inconsistent	Incorrect cuff fit or posture	Review Section 4.4 and 6.3
IHB indicator appears	Irregular pulse detected	Consult a healthcare provider
Bluetooth not connecting	Device not in pairing mode	Hold MEMORY 5 sec to enter pairing mode

8. Error Codes

The following table provides a complete reference for all error codes displayed by the CT-200. Codes E1–E4 are also described in Section 7.4. Codes E5–E9 are device-internal and will require service centre attention.

Code	Description	Auto-Recovery	Service Required
E1	Cuff not detected	No	No
E2	Motion artifact / measurement cancelled	Yes	No
E3	Over-pressurisation (>300 mmHg); auto-deflate	Yes	No (if isolated)
E4	Cuff displacement detected	No	No
E5	Pressure sensor failure	No	Yes
E6	Oscillometric algorithm timeout	No	Yes
E7	Memory read/write failure	No	Yes
E8	Bluetooth module error	Partial	No (restart)
E9	Internal clock failure	No	Yes

8.1 Error Codes During Startup

If an error code is displayed immediately upon power-on, it typically indicates a persistent hardware fault. Codes E5, E7, and E9 displayed at startup require mandatory service centre inspection before further use. Do not attempt to use the device if any of these codes appear at startup.

9. Technical Specifications

9.1 Measurement Method

Oscillometric method, fully automated, upper arm.

9.2 Measurement Range

Blood Pressure: 20–280 mmHg. Pulse Rate: 40–199 bpm.

9.3 Accuracy

±3 mmHg (blood pressure); ±5% of reading (pulse rate). See Section 6 for clinical validation details.

9.4 Power Supply

4 × AA/LR6 alkaline batteries (included), or DC 6V/1A via AC adapter (included). Battery life: approximately 300 measurements. Low-battery warning displayed when approximately 30 measurements remain.

9.5 Environmental Conditions

Condition	Operating	Storage/Transport
Temperature	10°C to 40°C	–20°C to 55°C
Relative Humidity	15% to 85% RH (non-condensing)	10% to 95% RH (non-condensing)
Atmospheric Pressure	700–1060 hPa	500–1060 hPa

9.6 Physical Characteristics

Main unit: 135 × 95 × 55 mm, 280 g (without batteries). Standard cuff: 22–42 cm arm circumference. Tube length: 60 cm.

9.7 Wireless Interface

Bluetooth 5.0 LE, HDP profile. Range: up to 10 m line-of-sight. USB: micro-USB, USB 2.0 Full-Speed.

10. Regulatory and Compliance Information

10.1 Regulatory Approvals

The CT-200 has received the following regulatory approvals:

- CE Mark (MDR 2017/745, Class IIa)
- FDA 510(k) clearance (K-number: K240987)
- Health Canada Class II Medical Device Licence
- TGA (Australia) ARTG Entry

10.2 Standards Compliance

The device has been designed and tested to comply with the following standards:

- IEC 60601-1:2005+AMD1:2012+AMD2:2020 (General requirements for basic safety and essential performance)
- IEC 60601-1-2:2014+AMD1:2020 (Electromagnetic compatibility)
- IEC 60601-2-30:2009+AMD1:2013 (Automated non-invasive sphygmomanometers)
- ISO 81060-2:2018 (Clinical validation)
- ISO 15223-1:2021 (Medical device symbols)

10.3 Manufacturer Information

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Appendix A: Glossary

Term	Definition
Systolic BP	The peak pressure in the arteries during a heartbeat (measured in mmHg).
Diastolic BP	The pressure in the arteries between heartbeats (measured in mmHg).
Oscillometric	A measurement method that detects oscillations in arterial wall movement caused by blood flow.
IHB	Irregular Heartbeat indicator — flagged when the device detects rhythm irregularity.
mmHg	Millimetres of mercury — the standard unit for blood pressure measurement.
ISO 81060-2	International standard specifying requirements for clinical validation of automated BP monitors.
EMC	Electromagnetic Compatibility — the ability of a device to function correctly in its electromagnetic envir
HDP	Health Device Profile — a Bluetooth profile for medical device data exchange.